

Toward Testing External Gravity Cosmology: Data and Research Directions

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Abstract

External Gravity Cosmology (EGC) proposes that the apparent acceleration and anisotropy of our universe can be explained not by dark energy or a primordial singularity, but by gravitational forces from massive structures beyond our observable horizon. This idea, if correct, would overturn the foundations of modern cosmology. Such a sweeping proposal cannot be advanced by a single individual, but requires a distributed program of collaboration.

This document outlines a roadmap of testable directions. It highlights opportunities to reanalyze existing data from Planck, Pantheon+, SDSS, and JWST, to mine large-scale structure surveys such as DESI and Euclid, and to anticipate gravitational-wave insights from LISA. Each project is formulated as an entry point for researchers who wish to probe the predictions of EGC. The aim is not to provide final answers, but to open the door for systematic testing of a bold new framework.

1 Introduction

External Gravity Cosmology (EGC) challenges the deepest assumptions of modern cosmology. Instead of invoking dark energy, inflation, or a primordial singularity, EGC proposes that the apparent acceleration of the universe and many of its observed anomalies can be explained by gravitational influences from massive structures beyond our observable horizon. In this framework, the cosmos we see—Our Universe (OU)—is a finite patch embedded within a larger, anisotropic gravitational landscape, The Universe (TU). Expansion, redshift, and even the texture of the cosmic microwave background (CMB) are not relics of a single explosive origin, but real-time signatures of external gravitational wells shaping our view from the inside.

This is a groundbreaking proposal. To properly assess it requires more than one perspective, more than one dataset, and more than one researcher. The work ahead is not about a single calculation or a single paper—it is about building a program of inquiry that invites collaboration from astronomers, cosmologists, data scientists, and theorists who are willing to test bold ideas against the best observations we have.

Fortunately, we live in a data-rich era. The James Webb Space Telescope (JWST) is revealing unexpectedly mature galaxies and black holes at extreme redshifts. The Planck satellite has mapped the CMB with exquisite precision, highlighting anomalies such as the “Axis of Evil” that remain unexplained. Surveys like DESI are releasing unprecedented maps of large-scale structure, and future missions such as LISA will probe the gravitational-wave background. These are exactly the kinds of datasets that can be reinterpreted through the lens of EGC.

What follows is not an exhaustive research plan, but a non-exhaustive list of projects and directions where EGC makes specific, testable predictions. Each represents an opportunity for

collaboration and discovery. Together, they form the foundation for a new kind of cosmological program: one that seeks to map the invisible gravitational landscape beyond our horizon.

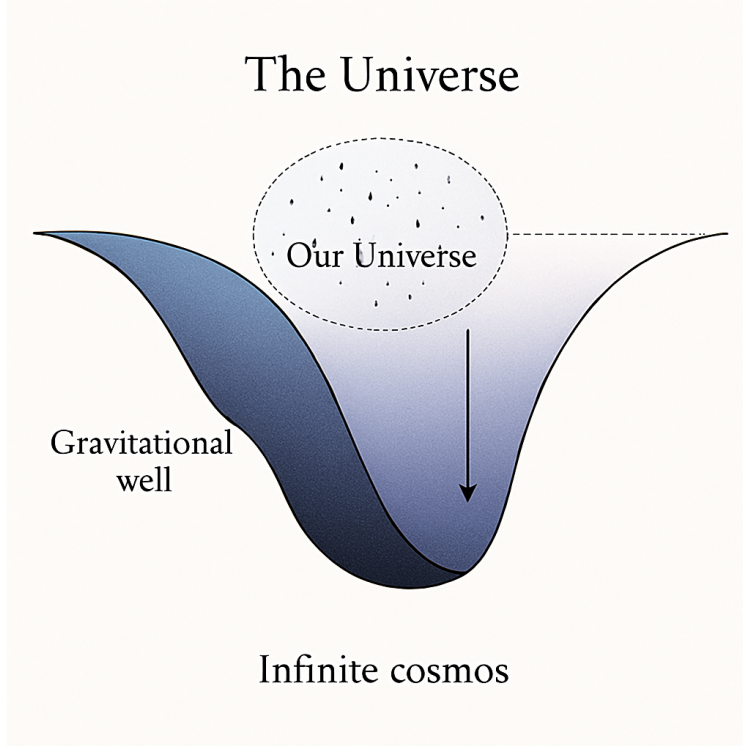


Figure 1: Schematic of Our Universe (OU) embedded within a larger anisotropic gravitational landscape (TU). External gravitational wells beyond the universal event horizon (OUEH) exert directional influences on expansion.

2 Proposed Research Directions

2.1 CMB and Redshift Correlations

A central prediction of External Gravity Cosmology is that anisotropies in the cosmic microwave background (CMB) and fluctuations in redshift surveys are not independent. Instead, they are two reflections of the same underlying gravitational landscape beyond our horizon.

This claim can be probed directly with existing datasets. The Planck satellite’s CMB temperature and polarization maps provide the highest-precision view of anisotropies on large scales. Meanwhile, supernova compilations such as Pantheon+ and galaxy redshift surveys from DESI and Euclid offer detailed measurements of how expansion rates vary across the sky. By comparing residuals in redshift–distance relations with CMB temperature fluctuations in the same directions, one could test whether the predicted alignment exists.

Such a correlation, if robustly established, would be revolutionary: it would mean that the CMB is not simply a relic of the Big Bang but a dynamic imprint of the gravitational texture of the larger universe. This project would likely require careful statistical cross-correlation techniques, but the data are already in hand.

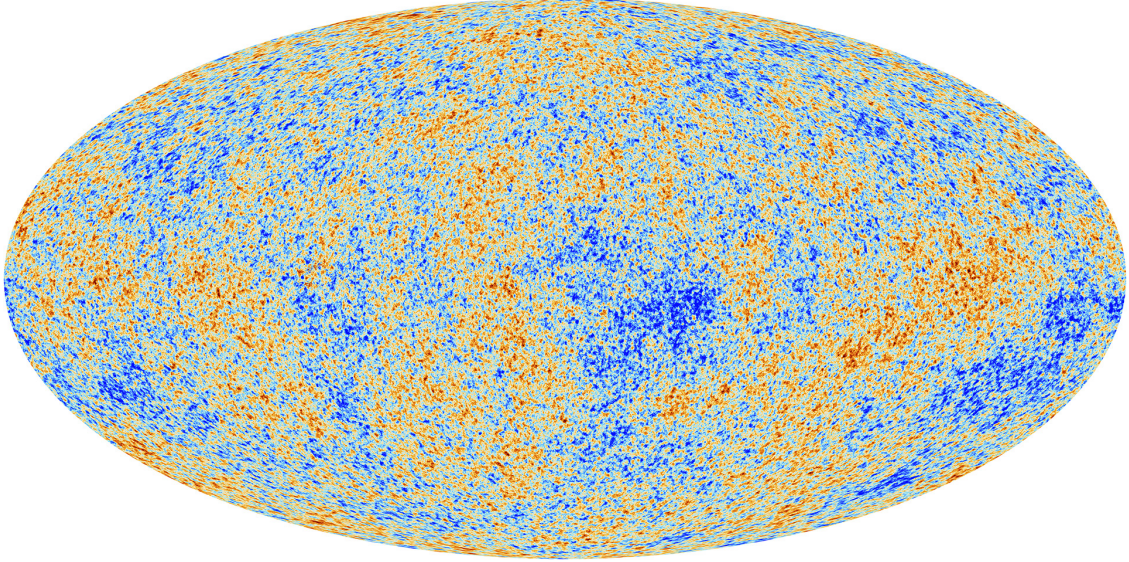


Figure 2: CMB temperature anisotropies measured by the Planck satellite. EGC predicts that these fluctuations should correlate with directional variations in redshift residuals from supernovae and galaxy surveys. Image credit: ESA/Planck Collaboration.

2.2 Supernova Data Reinterpretation

Type Ia supernovae are the “standard candles” that led to the discovery of accelerated cosmic expansion. In the Λ CDM framework, their dimming with distance is interpreted as evidence for dark energy. Under EGC, however, the same trend could emerge from gravitational redshift across large-scale potentials.

Reanalyzing supernova datasets such as Pantheon+ or the newer JWST time-delay lensed supernovae (e.g. SN H0pe) within a gravitational-redshift framework could test this idea. The methodology would involve fitting distance–redshift relations not with the standard Friedmann equations, but with alternative relations derived from gravitational potentials $\Phi(r)$. Bayesian model comparison could then assess whether the gravitational framework fits the data as well as, or better than, dark energy models.

If even partial consistency is found, this would be a major step forward: it would mean that the case for dark energy is not as ironclad as usually assumed, and that gravity itself may suffice to explain cosmic acceleration.

2.3 Quasar Redshift Anomalies

Quasars have long been a source of controversy in cosmology, with some studies reporting associations between high-redshift quasars and low-redshift galaxies, or discrete “quantized” redshift patterns. While these findings remain debated, EGC offers a natural lens through which to revisit them: if redshift has both metric and gravitational components, then unusual associations or discrete structures may be expected.

Datasets such as the Sloan Digital Sky Survey (SDSS) and the CatWISE2020 quasar catalog contain millions of quasars with precise redshifts and sky positions. Mining these catalogs for clustering anomalies, quantization signals, or alignments with CMB features could provide a fertile testing ground for EGC. Machine learning methods—already used in astrophysical anomaly

detection—could play a role here.

This line of work would appeal to data-driven collaborators. Even if many claimed anomalies vanish under rigorous analysis, showing where EGC does or does not align with the quasar record would itself be a valuable step toward clarifying the theory’s empirical reach.

2.4 Baryon Acoustic Oscillations (BAO)

In the standard cosmological model, the characteristic 500-million-light-year spacing of galaxies—the BAO scale—is understood as a fossil imprint of sound waves in the hot plasma of the early universe. EGC offers an alternative: these large-scale ripples could instead be signatures of gravitational structuring from mass concentrations beyond our horizon.

Testing this requires comparing galaxy clustering data with models of external gravitational influence. Current and upcoming datasets are well-suited: the Dark Energy Spectroscopic Instrument (DESI) is mapping tens of millions of galaxies, while Euclid will soon release precision 3D maps of the cosmic web. Simulations could explore whether external wells naturally produce quasi-regular spacing patterns, and whether they align with observed BAO features.

If BAO can be reframed as an external-structure effect, it would remove one of the strongest “early-universe” arguments for the Big Bang paradigm. Even partial reproduction of the BAO signal would make EGC far more compelling to data-oriented researchers.

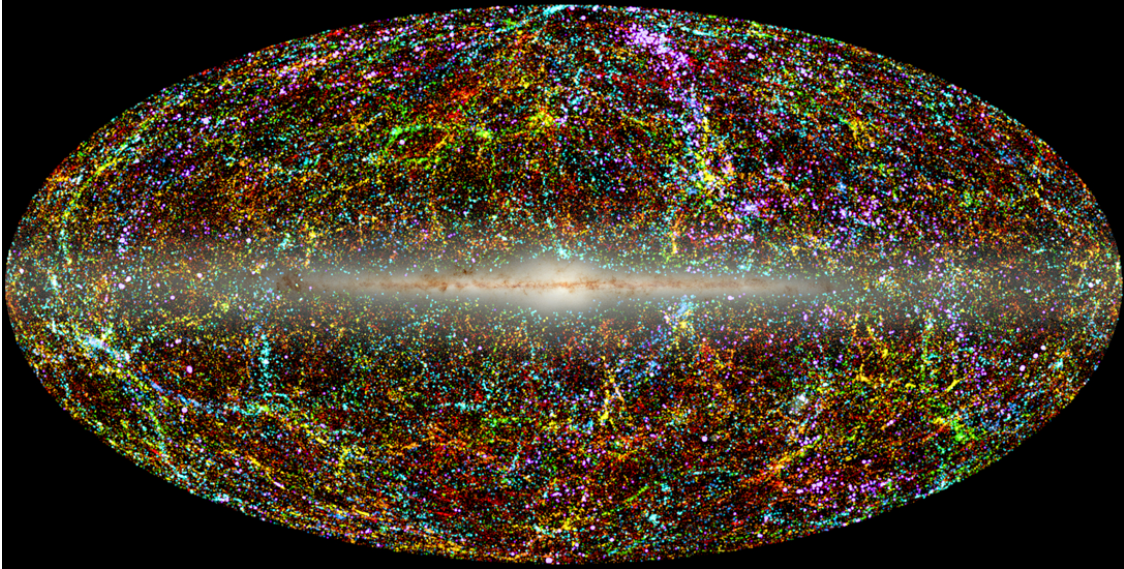


Figure 3: Distribution of nearby galaxies from the WISE and 2MASS surveys. Large-scale clustering patterns such as BAO are typically attributed to relic sound waves from the early universe. EGC offers an alternative interpretation: quasi-regular spacing driven by external gravitational structuring. Image credit: WISE/2MASS (Jarrett et al.).

2.5 JWST Early Galaxies and Black Holes

One of the most striking recent findings in cosmology comes from the James Webb Space Telescope (JWST). Within its first year, JWST revealed galaxies at redshifts $z > 10$ that appear far more massive and chemically mature than expected so soon after the Big Bang. Similarly, supermassive

black holes of $\sim 10^9 M_\odot$ have been found at redshifts beyond 6, straining models of black hole seed formation.

EGC provides a natural context for these anomalies: if the universe is infinite and eternal, then galaxies and black holes have had arbitrarily long timescales to evolve. The appearance of mature systems at high redshift is not an anomaly but an expected outcome of observing different “patches” of a much larger cosmos.

Future JWST deep fields and spectroscopic surveys will expand the sample of high- z galaxies. Cross-referencing these findings with the gravitational-redshift framework could yield testable signatures—such as whether the apparent ages and metallicities of early galaxies align better with EGC than with accelerated hierarchical formation under Λ CDM.

2.6 Gravitational Wave Signatures

If our observable universe is embedded in a lumpy gravitational environment, this may not only leave traces in redshifts and the CMB but also in the background of gravitational waves. Directional anisotropies in the stochastic gravitational-wave background could provide another window into external structures.

The upcoming Laser Interferometer Space Antenna (LISA), planned for launch in the 2030s, will probe gravitational waves at milli-Hz frequencies, complementing ground-based detectors like LIGO/Virgo/KAGRA. If EGC is correct, LISA could detect directional variations correlated with CMB anomalies or large-scale structure features.

While this is a longer-term project, including it in the roadmap signals that EGC is not just about reinterpretation of old data, but also about making novel predictions testable by the next generation of observatories.

3 Discussion

The projects outlined above vary in scope and technical demand, but they share a unifying feature: each points to a testable consequence of External Gravity Cosmology. Unlike many alternative models that remain purely speculative, EGC can be confronted with real data from ongoing and upcoming missions.

This is both a challenge and an opportunity. The challenge is that no single researcher can realistically reanalyze the Planck maps, fit supernova data, mine quasar catalogs, simulate BAO, interpret JWST deep fields, and model gravitational waves. It will require a distributed effort across specialties: observational astronomers, cosmological data analysts, computational modelers, and theorists alike.

The opportunity is that EGC offers a new rallying point. Instead of patching the standard model with increasingly exotic parameters, it calls us to rethink gravity itself as the primary architect of cosmic structure. It is precisely because this program is so ambitious that collaboration is essential. Each bullet point in the roadmap can be taken up as a standalone research project, yielding partial results that collectively test the theory.

The tools exist. The data are already streaming in. The question is whether the community has the curiosity and courage to pursue them.

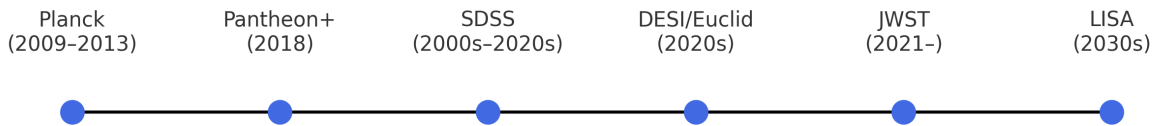


Figure 4: A roadmap of datasets relevant to testing EGC: past (Planck), current (Pantheon+, SDSS, DESI, Euclid, JWST), and future (LISA). Each provides opportunities to probe gravitational influences beyond our horizon.

4 Conclusion

External Gravity Cosmology reframes our place in the universe. It suggests that what we see—the accelerating expansion, the anisotropies in the CMB, the maturity of early galaxies—are not relics of a singular origin, but real-time signatures of a larger gravitational environment beyond our horizon.

This is a bold conjecture, and bold conjectures demand rigorous refutation. That is why the roadmap outlined here is so important: it provides clear directions for testing EGC against the strongest datasets we have—Planck, DESI, JWST, SDSS, Pantheon+, and eventually LISA.

No single study will settle the question. But taken together, these efforts can build a body of evidence for or against the idea that gravity alone, without dark energy or inflation, shapes the cosmos. Whether EGC ultimately stands or falls, the act of testing it will sharpen our understanding of the universe we inhabit.

We therefore extend an open invitation: to cosmologists, astronomers, data scientists, and all those who believe science advances through bold exploration. Join in mapping the unseen gravitational landscape beyond our horizon. The universe is bigger than we think—let’s find out how.

“Science advances by bold conjecture and careful refutation.”